

**Spatial diagonal weight**  $W = \text{diag}(w_1, \dots, w_N)$ ,  $0 < w_0 \leq w_i \leq w_1$

solution-preserving;  
stability constant unchanged

frozen spectrum: directions  
unreachable, scale bounded

the only route into  
the one-step update

**Channel  $S$  | zeroth-order: solution / fit**

zero-residual set, global minimizer,  
residual-error bridge;  
independent of  $W$  for  $w > 0$

**Level  $R$  | frozen linearization  
(not an independent channel)**

$H_{\text{lin}} = J^\top W J$  and its spectrum;  
acts only indirectly via  
multi-step gradient evolution

**Channel  $\theta$  | first-order: algorithm**

$g_w = J^\top W r$ , iteration trajectory,  
stationary points and  
basins of attraction

Level  $R$  acts on channel  $\theta$  only indirectly: within frozen training windows, via multi-step  $g_w$  evolution (dashed arrows)

**Controlled remainder  $E_{\text{rem}}$  (not a fourth channel)**

residual curvature  $H_{nl} = \sum_i w_i r_i \nabla^2 r_i$ , 3rd+-order Taylor terms, in-window curvature drift,  
preconditioner variation, exogenous jumps  
—— only explicit, controlled upper bounds